

How Students' Self-Regulated Learning Abilities Influence Intents and Engagement Goals in Chatbot-Assisted Writing

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Abstract

This paper examines how students' self-regulated learning (SRL) abilities influence their intents and engagement goals during chatbot-assisted argumentative writing. Through a systematic analysis of 229 conversation logs from 40 students, 15 distinct intents were identified and categorized into five engagement goals. The findings reveal that students with higher SRL abilities engage more actively with chatbots, particularly through the "Discuss" engagement goal, which involves seeking feedback and alternative perspectives. While the "Find" goal was the most frequently used across participants, its frequency also correlated positively with SRL abilities, highlighting a preference for active engagement. In contrast, the passive "Piggyback" goal was consistently employed regardless of SRL abilities, suggesting strategic use for convenience or cognitive load reduction. These results offer insights into the interplay between SRL abilities and chatbot interactions, providing a foundation for designing educational tools that foster active engagement and support diverse learning strategies.

CCS Concepts

• **Human-centered computing** → **Collaborative interaction**;
Web-based interaction; • **Applied computing** → **Education**.

Keywords

Chatbot-assisted writing, Intents, Engagement goals, Self-regulated learning

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1 Introduction

Argumentative writing is a cornerstone of education that fosters students' logical reasoning and argumentation skills [9]. Recent advancements in Large Language Model (LLM)-based chatbots have brought transformative potential to writing environments, offering students personalized feedback, real-time guidance, and support in crafting argumentative essays [4, 19]. By engaging with students during the writing process, these chatbots provide tailored suggestions and constructive feedback, helping students refine their ideas and structure persuasive arguments [5, 13, 15]. The integration of LLM-based chatbots into argumentative writing tasks is expected to enhance both the learning process and outcomes, establishing their potential as indispensable tools in education.

However, the effectiveness of LLM-based chatbots in fostering argumentation skills is significantly influenced by how students use these tools [10]. Students' intents—such as seeking guidance for writing, refining their essay expression, or challenging chatbot responses—and their engagement goals—such as to talk about a subject with chatbot or to use response that generated by chatbot—shape their interaction with chatbots [5, 6]. Active engagement involves students proactively seeking opinions from chatbots, collaboratively refining their writing, questioning their responses, resulting in deeper understanding and enhanced argumentation skills [10]. Conversely, passive engagement occurs when students uncritically accept chatbot-generated drafts as final without further thought or cognitive effort, which can impede meaningful learning [3, 4, 8]. Recognizing these intents and engagement goals is crucial, as chatbots can either empower students to develop stronger argumentation skills or foster dependency and superficial learning habits depending on how they are utilized [19].

To truly understand students' intents and engagement goals when using chatbots for argumentative writing, it is essential to go beyond traditional survey-based approaches. While prior research has often inferred engagement from self-reported data, this method lacks the depth needed to capture the nuances of real interactions [5]. Analyzing conversation logs from chatbot interactions offers a more objective and detailed understanding of how students use

these tools [18]. Conducting qualitative studies where researchers closely examine these logs with students, identifying the intents behind their interactions and grouping them into overarching engagement goals, is necessary for gaining actionable insights [10, 17]. Despite its importance, such in-depth research remains scarce. This study seeks to fill that gap by defining and categorizing engagement patterns through detailed log analysis, providing a richer foundation for optimizing chatbot-assisted learning environments.

Students' self-regulating learning (SRL) abilities may also play a pivotal role in shaping their intents and engagement goals with chatbots [13]. SRL abilities involve actively monitoring and controlling one's learning processes, including goal setting, self-evaluation, and strategy adjustment [24]. Students with stronger SRL abilities are more likely to engage actively with chatbots, whereas those with weaker SRL abilities may depend on chatbots for quick solutions, bypassing deeper cognitive engagement. Despite the importance of understanding this relationship, there is limited research exploring how SRL abilities influence students' intents and engagement goals with chatbots. Addressing this gap is essential for designing effective chatbot-assisted learning environments that promote active engagement [11, 21].

This study investigates the intents and engagement goals of students using LLM-based chatbots for argumentative writing and examines how these factors are shaped by and influence their SRL abilities. To achieve this, conversation log data from 40 students interacting with chatbots during argumentative writing tasks were analyzed to identify specific intents and categorize them into broader engagement goals. Semi-structured interviews were conducted to understand the rationale behind these intents, and affinity diagram analysis was used to classify engagement goals based on patterns observed in the logs. Additionally, a survey was administered to evaluate students' SRL abilities and explore how these abilities interact with their intents and engagement goals. This research aims to provide actionable insights into students' engagement behaviors and inform the design of chatbot-assisted learning environments that encourage critical and active engagement. The study is guided by the following research questions:

- RQ1: What are students' intents and engagement goals when using chatbots for argumentative writing?
- RQ2: How do students' SRL abilities influence their engagement goals when interacting with chatbots for argumentative writing?

2 Related work

2.1 Students' intents and engagement goals in chatbot-assisted writing

Understanding students' intents and engagement goals is fundamental for optimizing chatbot-assisted learning environments [18, 19]. Intents represent students' underlying motivations during their interactions with chatbots, such as seeking guidance, refining arguments, or challenging responses. Engagement goals, on the other hand, describe broader interaction patterns, such as collaborating with the chatbot or passively accepting its suggestions. These factors play a pivotal role in determining the quality of learning outcomes [1, 5]. For example, students who actively engage by questioning chatbot responses and collaborating on ideas often

demonstrate stronger argumentation skills. Conversely, passive engagement—such as uncritically accepting chatbot-generated responses—can lead to superficial learning. Thus, identifying and addressing students' intents and engagement goals is crucial for fostering active learning and promoting meaningful educational experiences [17].

Despite its importance, prior research has faced limitations in effectively capturing students' intents and engagement goals. Most studies have relied heavily on self-reported methods, such as surveys and questionnaires, to infer engagement patterns. For instance, Gao et al. extended the ICAP framework to assess engagement levels in chatbot interactions through a self-reported questionnaire [5]. While informative, these methods are constrained by inherent biases in self-reported data, such as overestimation of effort or inaccuracies due to memory recall. Moreover, self-reported methods fail to capture the nuanced dynamics of real-time interactions, providing a limited understanding of students' engagement behaviors. These shortcomings highlight the need for methodologies that can objectively analyze students' interactions with chatbots [12].

To overcome these limitations, this study adopts an approach that integrates conversation log analysis with the Affinity Diagram technique. Conversation logs offer a reliable source of observable evidence, capturing real-time interactions between students and chatbots during argumentative writing tasks [18]. These logs allow researchers to infer students' intents based on identifiable patterns, such as requests for alternative perspectives or sentence rephrasing [6, 23]. Using the Affinity Diagram technique, these intents can then be systematically grouped into broader engagement goals, facilitating a deeper understanding of students' interaction behaviors [17]. This method addresses the limitations of traditional self-reported approaches by providing a more nuanced and objective analysis of students' engagement patterns [7].

To further enhance the accuracy of intent identification, this study combines conversation log analysis with semi-structured interviews. While log data provides critical insights into observable behaviors, semi-structured interviews uncover students' intrinsic motivations by enabling researchers to explore their rationale for specific actions. The flexibility of semi-structured interviews allows for follow-up questions that delve deeper into the intents behind chatbot interactions [16]. By triangulating findings from conversation logs with interview data, this approach ensures a comprehensive understanding of students' intents and engagement goals. This combined methodology addresses the limitations of existing research and provides actionable insights for designing chatbot-assisted learning environments that promote active engagement.

2.2 The role of students' self-regulated learning in chatbot-assisted writing

Understanding students' SRL abilities is essential for interpreting how they engage with chatbots and benefit from their use. SRL refers to learners' capacity to actively manage their thoughts, emotions, and behaviors to achieve learning goals. This process is particularly relevant in chatbot-assisted educational environments, where students independently decide how to interact with the tool. Key SRL abilities include text revising, where students refine their

writing based on feedback, and peer learning, which involves collaboratively exploring chatbot suggestions for deeper understanding. Additionally, SRL abilities manifest through feedback handling, where students actively engage with chatbot-generated suggestions to improve their arguments, and interest enhancement, as students use chatbots to explore diverse perspectives that maintain their motivation and curiosity. Motivational self-talk, or using chatbots to support self-encouragement and goal reinforcement, further reflects SRL in action. Each of these strategies highlights how SRL abilities influence students' intents and engagement goals during chatbot interactions.

Despite the apparent relevance of SRL abilities in shaping students' chatbot interactions, research exploring this relationship remains scarce. Limited studies have examined how students' SRL abilities affect their intents (e.g., seeking feedback or rephrasing sentences) and engagement goals (e.g., active collaboration or passive reliance on chatbot responses) [14]. Understanding this connection is essential for identifying factors that contribute to passive engagement and designing interventions to encourage active learning. For example, students with lower SRL abilities may depend on chatbots for quick solutions, bypassing critical cognitive processes, while those with higher SRL abilities may actively leverage chatbot feedback to refine their work. This study addresses the gap by investigating how students' SRL abilities influence their intents and engagement goals when using chatbots. A widely-used SRL assessment survey was employed to evaluate these strategies, providing a lens to interpret students' behaviors in chatbot-assisted writing environments.

3 Methods

3.1 Participants

This study recruited 40 participants, comprising both undergraduate and graduate students aged between 22 and 32 years ($M=25.5$, $SD=1.7$), via an online bulletin board. To mitigate potential biases associated with prior experience or knowledge, participants were selected from approximately 20 different academic majors, ensuring a diverse sample. All participants reported using chatbots at least once per week for writing purposes and had engaged in at least one instance of argumentative writing within the past year. Each participant utilized their personal laptop for the experiment. As a token of appreciation, participants who completed the entire experiment received a \$10 coffee voucher. Ethical approval for this study was obtained from the Research Ethics Committee of Seoul National University of Science and Technology.

3.2 Chatbot-assisted writing platform

A web-based chatbot-assisted writing platform was developed to provide students with an experience closely resembling the real-world use of chatbots for chatbot-assisted writing. Built using a Node.js-based framework, the platform ensured seamless compatibility across Mac, Windows, and Android devices. It featured a text editor for students to compose argumentative essays and a chatroom integrated with a GPT-4o API chatbot. Students were able to interact with the chatbot through conversations, like the existing ChatGPT. This design was intended to encourage natural interactions in a manner familiar to them. The chatbot supported

students throughout the writing process by offering real-time feedback, rephrasing suggestions, and alternative perspectives. This integration allowed students to engage actively with the chatbot, refining their arguments and improving their essays while closely simulating authentic chatbot-assisted writing scenarios.

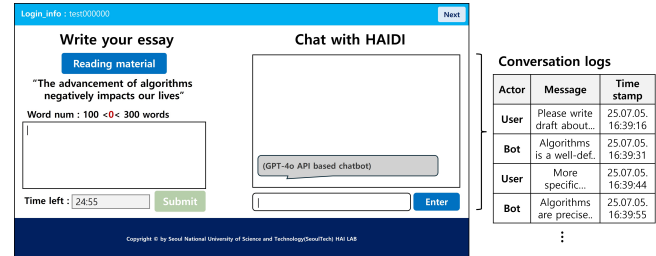


Figure 1: Graphic user interface of the chatbot-assisted writing platform. The right side displays an example of the conversation log being collected from participants when using platform.

All interactions between students and the chatbot, including conversations and essay submissions, were systematically recorded as conversation logs. These logs captured the detailed dynamics of how students collaborated with the chatbot during their writing tasks. To gain deeper insights into students' intents behind their interactions, the logs were further analyzed in combination with semi-structured interviews. This approach emphasized the chatbot's role in shaping students' engagement and provided a foundation for understanding their intents and learning behaviors.

3.3 Self-regulated learning ability assessment

To evaluate students' SRL abilities in the context of chatbot-assisted writing, we employed a modified version of the Writing Strategies for Self-Regulated Learning Questionnaire (WS-SRLQ), originally developed by Teng and Zhang [20, 22]. The modified WS-SRLQ measures five substrategies specific to writing: Text Revising, Peer Learning, Feedback Handling, Interest Enhancement, and Motivational Self-Talk, using a 7-point Likert scale ranging from (1 = "not at all true of me" to 7 = "very true of me").

For this study, the WS-SRLQ was tailored to exclude Motivational Self-Talk, as this strategy is either not directly applicable or not easily observable in the context of chatbot-assisted writing. The exclusion aligns with the study's focus on SRL abilities that are explicitly influenced by chatbot interactions [20]. Students' SRL scores were calculated by averaging the summed scores from the remaining substrategies. While these scores do not provide a comprehensive representation of a student's overall SRL abilities, they offer a meaningful understanding of their SRL abilities within the specific experimental context [20]. This modified assessment allowed us to evaluate how students regulate their learning behaviors when interacting with chatbots in chatbot-assisted writing.

3.4 Experimental procedures

The experiment began with participants submitting a consent form that included basic demographic information, such as name and

age. Sessions were scheduled based on participants' availability and conducted either face-to-face or online. The experiment comprised four phases: practice, argumentative writing with a chatbot, SRL assessment, and semi-structured interviews. In the practice phase, participants responded to a prompt—"Do you agree or disagree with the introduction of AI in professional fields?"—using a reading material overview. They wrote a 100–300-word argumentative essay within 25 minutes to familiarize themselves with the platform and task structure.

In the argumentative writing with a chatbot phase, participants addressed a new prompt: "The advancement of algorithms negatively impacts our lives." The writing conditions (25-minute time limit and 100–300-word range) were the same, but this time they engaged with the chatbot to seek feedback, rephrase sentences, and explore perspectives. After writing, participants completed the WS-SRLQ questionnaire in the SRL assessment phase, capturing their SRL abilities. Lastly, participants participated in semi-structured interviews to discuss their intents and engagement with the chatbot, providing deeper insights into their interaction patterns. This procedure allowed for a systematic examination of students' SRL abilities, intents, and chatbot-assisted writing behaviors.

3.5 Data analysis

To analyze students' intents when using chatbots, semi-structured interviews were conducted. Researchers began by reviewing each student's conversation logs from the argumentative writing with a chatbot phase. During the review, students were asked to explain the intent behind specific interactions, providing context for their actions. Follow-up questions, such as "Was the chatbot's response helpful?" and "How did you use the chatbot's response?", were used to explore underlying motivations. The interviews continued until researchers were confident they had sufficiently captured each student's intents. These identified intents were then classified into broader engagement goals using the affinity diagram approach, which organizes and groups related ideas to uncover common themes [7, 17]. Two researchers independently grouped similar intents while considering engagement levels (Active vs. Passive) and iteratively refined the groupings through discussions until a consensus was reached. Finalized groups were labeled to represent students' overarching engagement goals.

To investigate the relationship between students' SRL abilities and their engagement goals, correlation analyses were conducted using students' SRL scores derived from the WS-SRLQ. The correlation analysis examined how SRL scores influenced frequency of engagement goals. Usage frequencies were assessed to identify patterns in the volume of chatbot use. Given the limited sample size ($n = 40$), Kendall's tau correlation was used to ensure robust and reliable results. This analysis provided insights into how SRL abilities shaped students' engagement behaviors and interaction patterns with the chatbot.

4 Results

4.1 Students' intents and engagement goals

A total of 229 conversation logs were collected from 40 students during the chatbot-assisted argumentative writing tasks. Table 1 presents the basic statistics of the collected data, including students'

Table 1: Basic Statistics on Collected Data, Including Conversation Log Frequencies and Measured SRL Scores.

Element	M	SD
Conversation log	5.50	2.90
SRL score	5.31	0.88

conversation log frequencies and measured SRL scores. Of these, 29 conversation logs were excluded during preprocessing, as they contained dialogues that were not relevant to the learning context (e.g., "Hello?", "What's your name?"). Additionally, some conversation logs exhibited multiple intents and were consequently classified under all corresponding intents. As a result, a total of 220 conversation logs were labeled according to their identified intents. Table 2 presents the 15 identified intents and their corresponding five engagement goals, along with the frequency of each intent. Each engagement goal comprises two to four intents, and table 2 also includes annotations for the engagement level (Active or Passive) based on the extended ICAP framework [2, 5]. Additionally, examples of conversation logs for each engagement goal are provided in Appendix A.1.

The "Piggyback" goal was the only engagement goal categorized as Passive. Students with this goal tended to accept chatbot responses without deep reflection, primarily using the chatbot for convenience and efficiency. For example, intents under the Piggyback goal included writing easily and conveniently ($n = 9$), quickly modifying text into the desired format ($n = 21$), and incorporating structured details ($n = 11$). These behaviors highlight a reliance on the chatbot to simplify the writing process rather than actively engaging with its suggestions.

In contrast, the remaining four engagement goals—"Find," "Understand," "Instrumentalize," and "Discuss"—were categorized as Active, reflecting more meaningful interactions with the chatbot. Students pursuing the "Find" goal frequently searched for new Information, materials, or expressions ($n = 71$). The "Understand" goal involved grasping content more effectively, organizing thoughts, or improving conceptual understanding ($n = 21$). The "Instrumentalize" goal focused on learning how to use the chatbot effectively and prompting it strategically ($n = 7$). Lastly, the "Discuss" goal reflected efforts to engage in deeper dialogues with the chatbot, such as seeking alternative perspectives ($n = 16$), requesting feedback ($n = 4$), or countering points they found unconvincing ($n = 9$).

4.2 Effect of students' srl abilities on engagement goals with chatbot

The correlation analysis between students' SRL scores and the usage frequencies of engagement goals revealed important patterns in how students interacted with the chatbot. A representation of the engagement goal usage and total frequencies can be seen in Table 3, which illustrates the distribution of engagement behaviors and their relationship with SRL abilities. The total frequency of chatbot usage showed a significant positive correlation with SRL scores ($p < .05$), suggesting that students with higher SRL abilities interacted with the chatbot more frequently. Among the five engagement goals, "Find" had the highest average usage frequencies across

Table 2: Identified Student Intents with Frequencies and Organized Engagement Goals Annotated with Engagement Levels.

Engagement goal	Intent	N	Engagement level
Piggyback	1. To write easily and conveniently.	9	Passive
	2. To request improvements for enhancing writing quality.	22	
	3. To quickly modify the text into the desired format.	21	
	4. To write content that includes the desired structure and details.	11	
Find	5. To find appropriate expressions.	7	Active
	6. To gain knowledge for better writing.	4	
	7. To search for new materials and examples.	71	
Understand	8. To quickly grasp relevant information.	11	Active
	9. To better understand content and concepts.	21	
	10. To organize my thoughts.	4	
Instrumentalize	11. To learn how to use the chatbot effectively.	3	Active
	12. To prompt the chatbot.	7	
Discuss	13. To seek alternative perspectives.	16	Active
	14. To request feedback on my writing.	4	
	15. To counter or refute points I find unconvincing.	9	

Table 3: Correlation Analysis Between Students' SRL Scores and Frequencies of Engagement Goals.

Engagement goal	M (SD)	Kendall's τ	P value
Piggyback	1.57 (1.99)	0.04	.82
Find	2.05 (1.75)	0.33	.04*
Understand	0.9 (1.00)	-0.07	.66
Instrumentalize	0.25 (0.58)	-0.16	.31
Discuss	0.725 (1.19)	0.38	.01**
Total frequency	5.5 (2.90)	0.32	.04*

Note. M: Mean, SD: Standard Deviation, * $p < 0.05$, ** $p < 0.01$

students, indicating it was the most frequently employed strategy. Furthermore, the analysis showed significant correlation between SRL scores and the usage of the “Find” goal ($p < .05$), suggesting that students with higher SRL abilities may have required more information for gathering objective and unbiased viewpoints or for incorporating it into their essays.

Additionally, the “Discuss” engagement goal demonstrated a significant positive correlation with SRL scores ($p < .01$). Although it was not the most frequently used engagement goal (average usage frequency: 0.725), students with higher SRL scores were more likely to engage in discussions with the chatbot. This behavior included seeking alternative perspectives, requesting feedback, and countering points, highlighting the role of SRL abilities in fostering deeper, more meaningful interactions with the chatbot.

Other engagement goals, such as “Piggyback” and “Understand,” did not show significant relationships with SRL scores. Notably, the “Piggyback” goal, characterized by passive behaviors like quickly modifying or simplifying text, had the second-highest usage frequency but showed no significant association with SRL scores, reflecting its passive nature.

These findings emphasize that some engagement goals, like “Find,” are universally adopted by students, and goals such as “Find” or “Discuss” are more influenced by students' SRL abilities. This

highlights the importance of designing chatbot-assisted writing environments that encourage active engagement, especially for students with lower SRL abilities.

5 Discussion and conclusion

This study investigated students' intents and engagement goals while using a chatbot for argumentative writing. A total of 15 intents were identified and organized into 5 engagement goals, offering a detailed framework for understanding students' interactions. The findings revealed that students' SRL abilities significantly influenced their engagement patterns. In particular, students with higher SRL abilities demonstrated greater usage frequencies for the “Find” and “Discuss” engagement goal, which involved seeking more information and alternative perspectives, requesting feedback, and countering points. These results highlight the potential of engagement goals as a diagnostic tool for identifying students' SRL abilities and delivering timely, targeted interventions to improve learning outcomes. For instance, teachers can infer students' SRL abilities through engagement patterns without relying on traditional measurement methods like surveys, enabling them to focus more closely on students who need support. Engagement goals thus serve as a valuable lens for understanding and optimizing students' interactions with chatbots in argumentative writing.

Interestingly, passive engagement goals, such as “Piggyback,” were not influenced by students' SRL abilities. The “Piggyback” goal, characterized by tasks like quickly modifying text or writing with minimal effort, was utilized consistently by all students, regardless of their SRL abilities. This consistency suggests that even students with higher SRL abilities may employ passive engagement goals strategically—for instance, to reduce cognitive load or manage time effectively. However, all significant correlations with SRL abilities were observed in active engagement goals, such as “Find” and “Discuss.” This finding underscores the importance of fostering active engagement with chatbots to enhance learning outcomes, rather than attempting to regulate or reduce passive engagement behaviors. Encouraging active engagement may be a more effective

approach for helping students use chatbots more meaningfully in their learning processes.

These findings provide new insights into supporting the effective use of chatbots in real classrooms. Specifically, teachers can move beyond merely tracking the frequency of chatbot usage to analyzing students' intents and engagement goals in chatbot-assisted writing. Based on this understanding, they can provide more specific guidance and support, such as helping students refine their ideas or deepen their thinking through discussions with the chatbot. Additionally, our study has the potential to guide the design of educational chatbots that foster students' active learning.

While this study provides valuable insights into the relationship between students' SRL abilities and their engagement goals with chatbots, it is not without limitations. The small sample size of 40 participants may restrict the generalizability of the findings. Furthermore, the study was confined to argumentative writing, which may not capture the full spectrum of intents or engagement goals in other contexts or tasks. Future research should investigate these dynamics across diverse writing tasks, learning contexts, and larger sample sizes to develop a more comprehensive understanding of how students engage with chatbots in educational settings. Moreover, further exploration of the relationships between individual SRL sub-strategies and engagement goals could provide deeper insights into students' SRL abilities, informing the design of more tailored support systems. Additionally, analyzing the editing log of students' argumentative texts in chatbot-assisted writing could offer valuable insights into how these interactions shape writing development and SRL abilities.

Despite these limitations, this study contributes to the growing body of knowledge on chatbot-assisted writing by identifying and analyzing engagement goals influenced by SRL abilities. The findings offer practical implications for designing educational chatbots that encourage active engagement and support students in developing their SRL abilities. Ultimately, this research provides a meaningful direction for enhancing the effectiveness of chatbot-assisted writing environments and fostering productive interactions between students and AI tools.

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A Appendix

A.1 Questionnaire and Example Conversation Logs

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Table 4: Scale and Items of the Writing Strategies for Self-Regulated Learning Questionnaire (WS-SRLQ).

Scale	Item
Text revising	1. When revising with chatbot, I check whether my composition meets the expectations of the prompt. 2. When revising with chatbot, I check spelling and punctuation. 3. When revising with chatbot, I check the cohesiveness or connection among sentences. 4. When revising with chatbot, I check whether the topic and the content have been clearly expressed.
Peer learning	5. I received help during my writing time through brainstorming with the chatbot. 6. I discussed with the chatbot to gain more ideas for writing. 7. I perform writing tasks together with the chatbot.
Feedback handling	8. I am open to chatbots' feedback on my writing. 9. I try to improve my writing based on chatbots' feedback.
Interest enhancement	10. I try to connect the writing task with my personal interest. 11. I connect the writing task with my real life to intrigue me.

Table 5: Example Conversation Logs for Each Engagement Goal.

Engagement goal	Conversation log
Piggyback	"I need to write an essay on the <pasted argument>. Can you draft a version first?" "Write a three-paragraph essay on the issues of recommendation algorithms."
Find	"What are some cases of privacy violations caused by recommendation algorithms?" "Can you provide examples of companies that have effectively used recommendation algorithms?"
Understand	"Can you explain the following sentence in more detail? <pasted reading material>." "Please summarize the issues highlighted in <pasted reading material>."
Instrumentalize	"You are an essay writer. You are an AI that writes in a highly logical manner, and we will collaborate on writing an essay." "Hello, from now on, you are a writer specializing in essays and prose. Your current perspective is <pasted argument>."
Discuss	"Isn't this more of an issue with misinformation rather than a problem with recommendation algorithms?" "Where is the evidence supporting the claim that it reduces stress caused by information overload? If recommendations are too excessive, wouldn't they actually increase stress?"

Note. Original logs were collected in korean and translated into english